CSC 494 IoT Projects & Learning Ideas

Notice:

For general information about ASE course projects, read the ASE Project Stories in the ASE repository:

- https://github.com/nkuase/ASE/tree/main/ASE_story/project

In this slides, we only discss the specific projects in ASE 285 course projects.

CSC 494 Course Projects (Different from the Stories)

1. The Protypes are given:

The Project Skeleton Hardware & Software Prototypes are provided.

2. Students should choose two topics that they want to learn with AI tools.

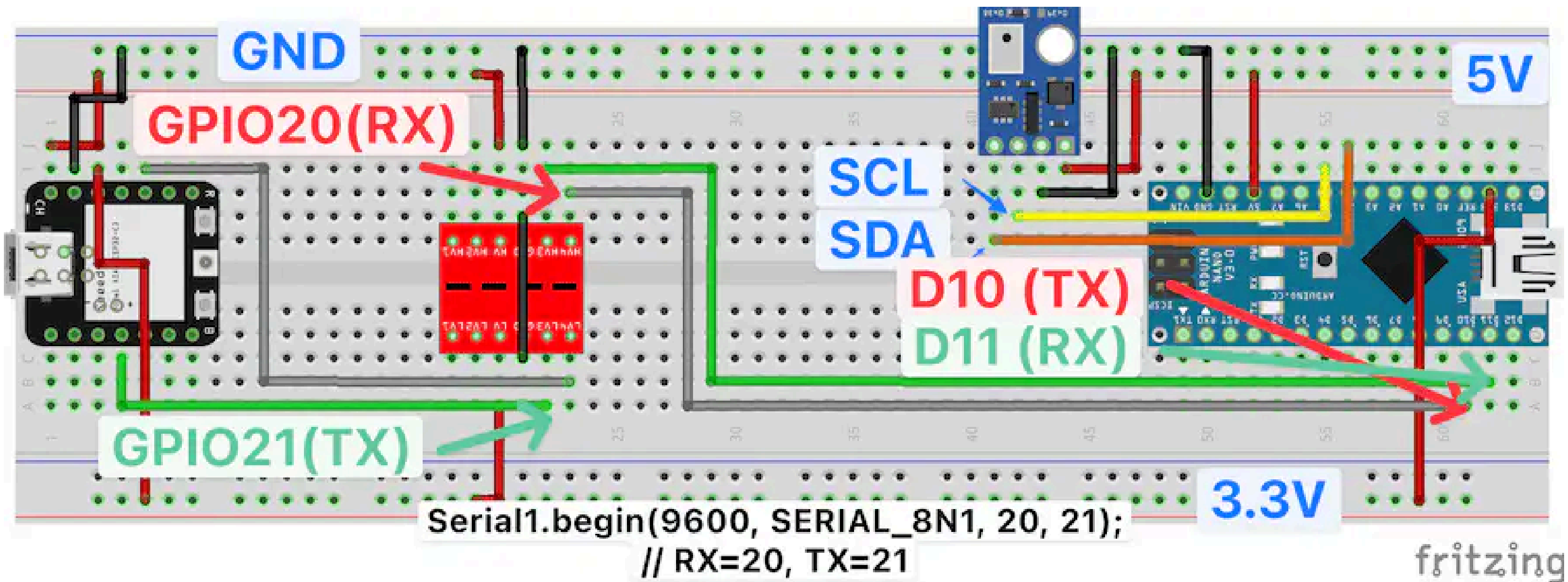
We will discuss the general tools & methodologies for using AI tools for learning any new topics.

Student Build IoT Project & Learn IoT Topics with AI Tools

1. In Stage 1: Students decide the IoT Project & two IoT Topics to Learn with AI Tools
2. In Stage 2: Make MVP of Iot Project & Learn one IoT Topic with AI Tools
3. In Stage 3: Make final version of IoT Project & Learn the 2nd IoT Topic with AI Tools

The Hardware Protype of IoT Project

The simplest IoT Project Hardware is as follows:



The Sensing of Temperature Data

1. The Temperature Sensor (TMP102) is connected to the Arduino Nano board.
2. The sensed temperature data is sent to the Ardunio Nano board via I2C communication protocol.

The Sharing of Temperature Data

3. The Ardunio Nano board sends the sensed temperature data to the ESP32C6 board via Serial communication protocol.
4. The ESP32C6 board shares the temperature data with any computer systems via BLE or WiFi communication protocol.
5. The ESP32C6 board uses 3.3V while the Ardunio Nano board uses 5V power supply, so the level shifter is needed between the two boards.

The ESP32C6 Board as a Wireless Network Gateway

6. The ESP32C6 board can act as a gateway of the IoT system.
7. Any computer systems (e.g., smartphone, laptop, desktop, etc.) can access the temperature data via WiFi communication protocol.

Possible Hardware Extensions of IoT Project

1. Communication Protocols: Use different communication protocols (e.g., BLE, LoRa, Zigbee, etc.) for sharing the temperature data.
2. Adding More Sensors: Add more sensors (e.g., humidity, light, motion, etc.) to the IoT system.
3. Adding Actuators: Add actuators (e.g., LED, motor, buzzer, etc.) to the IoT system.

Possible Software Extensions of IoT Project

1. Data Processing: Process the sensed data (e.g., filtering, averaging, etc.) before sharing it.
2. Data Visualization: Visualize the sensed data on a web page or mobile app.
3. Data Storage: Store the sensed data in a database for later analysis.

4. Data Analysis: Analyze the sensed data using machine learning algorithms.
5. Alerts: Send alerts (e.g., email, SMS, push notification, etc.) based on the sensed data.
6. Integration with Cloud Services: Integrate the IoT system with cloud services (e.g., AWS, Azure, Google Cloud, etc.) for scalability and reliability.

7. Security: Implement security measures (e.g., encryption, authentication, etc.) to protect the IoT system from cyber attacks.
8. Firmware Updates: Implement over-the-air firmware updates for the IoT devices.
9. Use different protocol stacks (e.g., MQTT, CoAP, HTTP, etc.) for communication between the IoT devices and computer systems.

Any IoT Project is Possible

1. Use low-power & high-efficiency ARM-based microcontrollers (e.g., Raspberry Pi Pico, STM32, etc.) for the IoT devices.
2. Add GPS module for location tracking.
3. Add robot & camera for moving surveillance system.

Is Your Imagaination the Limit?

So far, your imagination has been the limit, mainly because you don't know what you don't know.

Now, we are living in a totally different world.

1. Actually, there is no limit as you can discuss with **AI tools** to help you implement any IoT project ideas.
2. AI will teach you what you didn't know, and even help you learn and implement anything you want and need.
3. The only limit is your will & schedule to learn and implement your IoT project ideas. (Maybe budget is another limit, but we will try to minimize the cost of your IoT projects.)

Use AI Tools to Learn Anything You Want & Need

1. Choose two topics that you want/need to learn for your project.

- Communication systems (e.g., BLE, LoRa, Zigbee, etc.)
- Embedded systems (e.g., ARM-based microcontrollers, RTOS, etc.)

- ROS (Robot Operating System) (e.g., ROS2, navigation, etc.)
- IoT security (e.g., encryption, authentication, etc.)
- Cloud computing (e.g., AWS, Azure, Google Cloud, etc.)
- Data analysis (e.g., machine learning, data visualization, etc.)

...

2. Choose AI tools to help you learn the topics effectively.

- Build your 2nd Brain System with AI tools.
- Use the 2nd Brain System to know how you learn the best.

3. Share your learning process with your peers.
4. Learn from your peers' learning process.
5. Learn from the professor's experience of learning with AI tools.
6. Learn how to continuously learn anything you want & need with AI tools.